Inference of cytoskeletal geometry of cells from traction force microscopy

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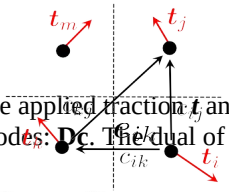
Introduction

Mechanical forces are recognized as key factor in cellular response. Therefore, the experimental measurement of these forces has an important role in numerous scientific fields in biology [1]. Traction Force Microscopy (TFM) has become a versatile and useful technique for determining the tractions exerted by cells on hydro gel substrates [2]. We here propose techniques for inferring contractile stress distribution, mechanically compatible with the measured tractions from TFM results [3].

Methods

We resort to two methods for inferring a i) contractility field **c** on a set of dipoles, and ii) a nodal stress field **σ**.

Dipoles approach: We construct a connectivity matrix **D** of mechanical dipoles with contractility compatible with TFM tractions . The contractility is deduced by solving a *regularized inverse least-square* problem for adapted **D**:



where $\boldsymbol{\xi}$ is the error between the applied traction $\boldsymbol{t}$ and the force of the dipoles at all nodes: $\mathbf{Dc}$. The dual of this problem,

$$\min_{\boldsymbol{\xi},\mu} \left(2\alpha\boldsymbol{\xi}^T\boldsymbol{t} + \alpha\|\boldsymbol{\xi}\|^2 + \|\mu\|^2\right)$$

$$\text{s.t.} \quad \boldsymbol{\mu} + \mathbf{D}^T\boldsymbol{\xi} \geq 0$$

is used to update the topology in **D** by removing dipoles when c_{ij}=0 and adding them when $(\boldsymbol{\xi}_i - \boldsymbol{\xi}_j)^T\mathbf{e}_{ij} < 0$.

Continuum approach: A piece-wise linear stress field is inferred which is in balance to the traction force field. The following least-square problem is solved.

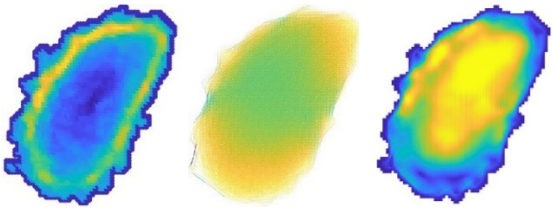
$$\min_{\boldsymbol{\sigma}} \left(\|\mathbf{B}\boldsymbol{\sigma} - \mathbf{f}\|^2 + \alpha\|\boldsymbol{\sigma}\|^2 + \beta\|\nabla\boldsymbol{\sigma}\|^2\right)$$

$$\text{s.t.} \quad \sigma_I, \sigma_{II} \geq 0$$

where σ_I and σ_{II} are the principal stresses. The objective aims at minimising mechanical imbalance after FE discretisation. Regularisation parameters α and β aim at reducing the stresses and oscillations in the solution.

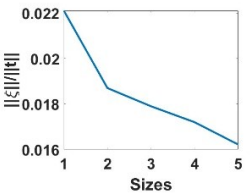
Results

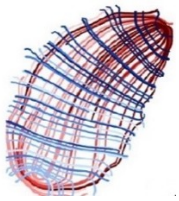
We tested our approach for simple toy problems for validating the values of the stress results. Figure 1 shows the results on an example experimentally measured traction fields. We resort elements with quadratic (Q8) stress interpolation for matching the linear traction field. Stress lines of principal stress are computed by integrating the stress trajectories [4**].**



a) b) Dipoles of size 1:5 c) ||dev(**σ**)||

Figure 1. Traction field, obtained dipoles of size 1 to 5, and norm of deviatoric stresses.





a) Relative nodal error evolution b) Stress lines (principal stresses)

Figure 2. Evolution of nodal error as a function of dipole size, and stress lines for cell in Fig 1a.

Conclusion

While the construction of the dipole structure is not fully satisfactory, the reconstruction of the stress lines are better reproduced. The error of the continuum approach (~1e-2) is comparable to the dipole nodal error. Current work focuses on a hybrid approach that combines dipoles and continuum approaches.

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References

1. Butler, JP et al, Cell Physiol. 282(3):595-605, 2002.
2. Godeau, AL et al, eLife. 28(11): e71032, 2022.
3. Soine, JRD et al, Comp Biol. 11(3): e10, 2015.
4. Wang, J et al, Comput. Methods Appl. Mech. Engrg, 452, 2026